From Barycentre Fibres to Secant Geometry

Positive tropicalisation of the posterior tower and its Berkovich factorisation

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October 13, 2026

Abstract

The companion paper [1] withdrew a proposed functor from finite stochastic kernels to Berkovich spaces. The proposal was ill-defined, because $\mathbb{R}$ does not sit isometrically inside a non-Archimedean field. This note builds the bridge that does exist and carries the tower's barycentre fibres to secant geometry. Probabilities are taken in a real closed non-Archimedean field K, such as the real Hahn or Puiseux series in a parameter t. They are then positive elements, and the valuation is a semiring homomorphism on the positive cone. Six results follow, each checked exactly.

1. The valuation is a morphism of monads, from the distribution monad over K to the normalised min-plus distribution monad $\mathsf{D}_{\mathbb{T}}$. The whole tower therefore maps to a tropical tower.
2. Every point of the tropical barycentre fibre lifts to an exact positive decomposition of the given distribution.
3. For any positively parametrised model V, the valuation image of its k-mixtures is exactly the tropical k-mixture of the positive tropicalisation of V. For toric models this is a tropical secant of a linear space.
4. On the skeleton of the Berkovich analytification, the analytified barycentre composed with the retraction equals the tropical barycentre. On positive rigid points this square commutes, and positivity is exactly what makes it commute.
5. The Bayes posterior tropicalises to the hard argmin assignment. This identifies the nearest-atom pooling of the earlier scripts as a *zero-temperature shadow*, and its Voronoi walls as tropical walls.
6. The component-collision "walls" of mixture geometry tropicalise to tie loci, where the weight-split segment becomes a pair of rays.

For the 3×3 and $2 \times 2 \times 2$ independence models, tropical and classical mixture dimensions agree for $k \leq 3$ and $k \leq 2$ respectively. Script `check_tropical.py` runs 8 checks, all passing.

1 Why a real non-Archimedean field

A non-Archimedean absolute value cannot restrict to the usual one on $\mathbb{R}$. So a probability vector over $\mathbb{R}$ has no Berkovich point in any meaningful sense, and every coordinate of it has valuation 0. To see geometry, probabilities must carry *scale*. Concretely, one works with families $p(t)$ whose coordinates behave like $c\,t^a$ as $t \to 0$. Such families are exactly elements of a field of generalised power series in t.

Definition 1.1. Let $K = \mathbb{R}((t^{\mathbb{R}}))$ be the field of real Hahn series, the formal sums $\sum_{a\in S} c_a t^a$ with $S \subset \mathbb{R}$ well ordered. It is real closed and complete for the valuation $\mathrm{val}(f) = \min S$, with value group $\mathbb{R}$. Write $|f| = e^{-\mathrm{val}\, f}$. f is *positive* if its leading coefficient $c_{\min S}$ is positive. The real Puiseux series (exponents in $\frac{1}{N}\mathbb{Z}$) form a subfield, on which all exact checks are carried out.

The single fact behind everything that follows is this:

$$f, g > 0 \implies \mathrm{val}(f+g) = \min(\mathrm{val}\, f, \mathrm{val}\, g), \qquad \mathrm{val}(fg) = \mathrm{val}\, f + \mathrm{val}\, g. \tag{1}$$

Leading coefficients of positive series cannot cancel. Hence $\mathrm{val}\colon (K_{>0}, +, \times) \to (\mathbb{R}, \min, +)$ is a homomorphism of semirings. This is Maslov dequantisation, and in statistical language it is the Laplace principle: $-\lim_{t\to 0} \log_t \sum_i e^{-a_i \log(1/t)} = \min_i a_i$.

Interpretation. A positive K-point is a one-parameter family of distributions together with its large-deviation rates. The valuation keeps the rates and forgets the prefactors. Everything below is a statement about rates, and it is exact, not asymptotic: it is an identity in K read off at the leading order.

2 The tropical tower

Definition 2.1 (Two monads)**.** For a set X:

- $\mathsf{D}_K X$ is the set of finitely supported $p\colon X \to K_{\geq 0}$ with $\sum_x p(x) = 1$, with Dirac unit and barycentre multiplication;
- $\mathsf{D}_{\mathbb{T}} X$ is the set of finitely supported $v\colon X \to \mathbb{R} \cup \{\infty\}$ (finite on finitely many points) with $\min_x v(x) = 0$, with unit $\eta_{\mathbb{T}}(x) = (0 \text{ at } x,\ \infty \text{ elsewhere})$ and multiplication

$$\mu_{\mathbb{T}}(V)(x) = \min_u \big(V(u) + u(x)\big).$$

$\mathsf{D}_{\mathbb{T}} X$ is the *tropical simplex*; for finite X it is $\{v \in (\mathbb{R} \cup \infty)^X : \min v = 0\}$.

Theorem 2.2 (Valuation is a monad morphism)**.** $\mathsf{D}_{\mathbb{T}}$ *is a monad, and* $\mathrm{val}_X\colon \mathsf{D}_K X \to \mathsf{D}_{\mathbb{T}} X$, $p \mapsto \mathrm{val} \circ p$*, is a morphism of monads:*

$$\mathrm{val} \circ \eta = \eta_{\mathbb{T}}, \qquad \mathrm{val}_X \circ \mu_X = \mu_{\mathbb{T},X} \circ \mathrm{val}_{\mathsf{D}_{\mathbb{T}} X} \circ \mathsf{D}_K(\mathrm{val}_X).$$

Proof. $\mathsf{D}_{\mathbb{T}}$ is the normalised distribution monad of the semiring $(\mathbb{R}\cup\infty, \min, +)$. The monad laws are the semiring laws, and normalisation is preserved because $\min_x \min_u (V(u)+u(x)) = \min_u V(u) = 0$. For the morphism, $\sum_x p(x) = 1$ with $p \geq 0$ gives $\min_x \mathrm{val}\, p(x) = 0$ by (1), so val_X lands in $\mathsf{D}_{\mathbb{T}} X$. The unit square is $\mathrm{val}(1) = 0$, $\mathrm{val}(0) = \infty$. The multiplication square is (1) applied to $\mu(\Pi)(x) = \sum_\rho \Pi(\rho)\rho(x)$. Atoms of Π with equal valuation are merged by $\mathsf{D}_K(\mathrm{val})$ with summed weights, and their valuation is again the minimum. □

Corollary 2.3 (The tropical tower)**.** val *maps a coherent K-context $(\rho, \Pi, \Omega, \Sigma)$ to a coherent tropical context: $\mu_{\mathbb{T}}(\mathrm{val}\,\Pi) = \mathrm{val}\,\rho$, $\mu_{\mathbb{T}}(\mathrm{val}\,\Omega) = \mathrm{val}\,\Pi$, and similarly for the shadow. The two split idempotents $e_\perp = \eta\mu$ and $e_\top = \mathsf{D}(\eta)\mu$ of [1, Prop. 5.1] map to their tropical counterparts. Pooling a labelled random measure on a cell (a conditional expectation, [1, §4]) maps to the tropical cell barycentre* $\min_j(\mathrm{val}\, p_j + \mathrm{val}\, Z_j) - \min_j \mathrm{val}\, p_j$.

Proof. Coherence and the idempotents are built from η and μ, which commute with val by theorem 2.2. A cell barycentre is μ applied to the normalised restriction of p to the cell, and normalisation subtracts the minimal valuation. □

Checks T1 and T2 verify the monad laws for $\mathsf{D}_{\mathbb{T}}$ and the multiplication square, exactly, on random positive Puiseux data.

3 Fibres

Definition 3.1 (Classical and tropical k-fibres)**.** For $\rho \in \mathsf{D}_K X$ with all coordinates positive, the *k-fibre* $F_\rho^{(k)}$ is the set of positive decompositions $\rho = \sum_{i=1}^k w_i r_i$, with $w \in \mathsf{D}_K[k]$, $r_i \in \mathsf{D}_K X$ and all $w_i > 0$. For $v \in \mathsf{D}_{\mathbb{T}} X$, the *tropical k-fibre* $F_{\mathbb{T},v}^{(k)}$ is the set of $(w, u_1, \dots, u_k)$ with $w \in \mathsf{D}_{\mathbb{T}}[k]$ finite, $u_i \in \mathsf{D}_{\mathbb{T}} X$, and

$$\min_i \big(w_i + u_i(x)\big) = v(x) \qquad (x \in X).$$

Theorem 3.2 (Fibre lifting)**.** $\operatorname{val}\big(F_\rho^{(k)}\big) = F_{\mathbb{T},\operatorname{val}\rho}^{(k)}$. *That is, the valuation maps the classical fibre* onto *the tropical fibre, and every tropical decomposition is the rate pattern of an exact positive decomposition of the given ρ.*

Proof. $\subseteq$ is theorem 2.2. For $\supseteq$, let $(w, u) \in F_{\mathbb{T},v}^{(k)}$ with $v = \operatorname{val}\rho$. For each x, let $\alpha(x) = \{i : w_i + u_i(x) = v(x)\} \neq \emptyset$ be the set of *attainers*. For $i \notin \alpha(x)$ choose any positive $q_i(x)$ with $\operatorname{val} q_i(x) = w_i + u_i(x)$; this valuation is $> v(x)$. Put $m(x) = \rho(x) - \sum_{i \notin \alpha(x)} q_i(x)$. Its leading term is that of $\rho(x)$, so $m(x) > 0$ and $\operatorname{val} m(x) = v(x)$. Split $m(x)$ equally among the attainers, $q_i(x) = m(x)/|\alpha(x)|$. Then $\sum_i q_i(x) = \rho(x)$ exactly. Set $w_i^K = \sum_x q_i(x)$ and $r_i = q_i / w_i^K$. By (1), $\operatorname{val} w_i^K = \min_x(w_i + u_i(x)) = w_i + \min u_i = w_i$, and $\operatorname{val} r_i(x) = u_i(x)$. □

Check T3 carries out this construction exactly on 300 random tropical fibre points, 83 of whose coordinates have ties, and verifies every identity.

Proposition 3.3 (Tropical fibres are polyhedral and carry the convex-order ends)**.** $F_{\mathbb{T},v}^{(k)}$ *is a finite union of polyhedra. The union is indexed by* assignment types $\alpha\colon X \to 2^{[k]} \setminus \{\emptyset\}$*, and the cell of type α is cut out by the equalities $w_i + u_i(x) = v(x)$ for $i \in \alpha(x)$ and the inequalities $>$ for $i \notin \alpha(x)$. The tropical images of the ends of the classical fibre [1, Prop. 5.2] are its two distinguished points: the one-component decomposition $e_\perp \mapsto (w = 0, u_1 = v)$, and the vertex decomposition $e_\top \mapsto (w_x = v(x), u_x = \eta_{\mathbb{T}}(x))$.*

Proof. Each defining condition is a min-plus equation, which is a union of polyhedra indexed by which terms attain the minimum. The two points are the images under val of δ_ρ and $\sum_x \rho(x)\delta_{\delta_x}$, using corollary 2.3. □

4 Secants

Definition 4.1. A model $V \subseteq \mathsf{D}_K X$ is *positively parametrised* if $V = \varphi(\Theta_{>0})$ for a map φ whose coordinates are polynomials (or ratios of such) with nonnegative integer coefficients, on a positive orthant $\Theta_{>0} = K_{>0}^d$. Its *positive tropicalisation* is $\operatorname{trop}^+ V = \varphi_{\mathbb{T}}(\mathbb{R}^d)$, where $\varphi_{\mathbb{T}}$ replaces $+$ by min and $\times$ by $+$. The *k-mixture* is $\operatorname{Mix}_k(V) = \{\sum_{i \le k} w_i r_i : w \in \mathsf{D}_K[k]_{>0}, r_i \in V\}$, the positive part of the k-th secant variety of the Zariski closure. Its tropical analogue is $\operatorname{Mix}_k^{\mathbb{T}}(L) = \{\min_i(w_i + u_i) : w \in \mathsf{D}_{\mathbb{T}}[k], u_i \in L\}$.

Theorem 4.2 (Mixtures tropicalise to tropical secants)**.** *For a positively parametrised V,*

$$\operatorname{val}\big(V\big) = \operatorname{trop}^+ V, \qquad \operatorname{val}\big(\operatorname{Mix}_k(V)\big) = \operatorname{Mix}_k^{\mathbb{T}}\big(\operatorname{trop}^+ V\big),$$

with exact equality of sets, not merely up to closure. If $V = V_A$ is the toric (log-linear) model of an integer design matrix A, then $\operatorname{trop}^+ V_A$ is the row space of A (normalised by $\min = 0$), a classical linear space, and $\operatorname{Mix}_k^{\mathbb{T}}(\operatorname{trop}^+ V_A)$ is the tropical k-th secant of a linear space in the sense of Develin [4].

Proof. For $\theta \in \Theta_{>0}$, (1) gives $\operatorname{val}\varphi(\theta) = \varphi_{\mathbb{T}}(\operatorname{val}\theta)$. Since val maps $\Theta_{>0}$ onto $\mathbb{R}^d$, the first identity follows. The second applies theorem 2.2 to $\sum_i w_i r_i$ and chooses w_i and θ_i with prescribed valuations. For V_A, $\varphi(\theta)_x = \prod_j \theta_j^{a_{jx}}$ up to normalisation, so $\varphi_{\mathbb{T}}(\vartheta) = A^\top \vartheta$ minus its minimum. □

Remark 4.3 (Positivity and the tropical secant problem)**.** Over an algebraically closed field, $\operatorname{trop}(\operatorname{Sec}_k)$ can be strictly larger than the tropical secant of $\operatorname{trop} V$, because cancellation in sums produces extra tropical points [4, 5]. Theorem 4.2 shows that on the positive part, which is the part relevant to probability, there is no discrepancy at all. Equivalently, the positive tropicalisation of the nonnegative-rank-$\le k$ locus is the Barvinok rank-$\le k$ locus [5]. For real semialgebraic sets, the image of the valuation is the logarithmic limit set of Alessandrini [2], whose dimension is at most the dimension of the set.

Example 4.4 (Dimensions)**.** Generic Jacobian ranks of the parametrisations (check T6, T7) give the following dimensions:

model	k	tropical $\dim \operatorname{Mix}_k^{\mathbb{T}}$	classical $\dim \operatorname{Mix}_k$ (affine cone)
3×3 independence	1, 2, 3	5, 8, 9	5, 8, 9
$2 \times 2 \times 2$ independence	1, 2	4, 8	4, 8

In each case the tropical dimension already certifies the classical dimension, in the spirit of Draisma's tropical lower bound [6]. The numbers are floating-point Jacobian ranks at random points, with tolerance 10^{-6}, on 40 points per case.

5 The Berkovich factorisation

The valuation of a K-point factors through the Berkovich analytification. This is the precise sense in which the map from fibres to secants "is Berkovich".

Setup. Let $\mathbb{A}^m_K$ be affine space with coordinates $z_1, \dots, z_m$. Its Berkovich analytification $(\mathbb{A}^m_K)^{\mathrm{an}}$ is the space of multiplicative seminorms on $K[z]$ extending $|\cdot|$ on K, with the weakest topology making $\xi \mapsto |f|_\xi$ continuous [3]. A K-point y gives the rigid point $|f|_y = |f(y)|$. On the torus $\mathbb{G}^m_m \subset \mathbb{A}^m$ there are two maps:

- the *skeleton section* $s\colon \mathbb{R}^m \to (\mathbb{G}^m_m)^{\mathrm{an}}$, sending v to the Gauss point $|\sum_m a_m z^m|_{s(v)} = \max_m |a_m| e^{-\langle m, v\rangle}$;
- the *retraction* $\tau\colon (\mathbb{G}^m_m)^{\mathrm{an}} \to \mathbb{R}^m$, $\tau(\xi) = (-\log|z_1|_\xi, \dots, -\log|z_m|_\xi)$.

They satisfy $\tau \circ s = \mathrm{id}$ and $\tau(y) = \operatorname{val}(y)$ on rigid points. By Payne's theorem [9], the analytification of a variety is the inverse limit of its tropicalisations over all toric embeddings, and τ is the projection to one of them. Jell, Scheiderer and Yu prove the real analogue for semialgebraic sets [7].

Theorem 5.1 (Berkovich square)**.** *Let $\Phi\colon \mathbb{A}^m \to \mathbb{A}^n$ be a polynomial map with integer coefficients, for instance the k-mixture map $(w, r) \mapsto (\sum_i w_i r_i(x))_x$. Assume the residue field has characteristic 0 (true for $K = \mathbb{R}((t^{\mathbb{R}}))$). Then:*

(a) on the skeleton, $\tau \circ \Phi^{\mathrm{an}} \circ s = \Phi_{\mathbb{T}}$, where $\Phi_{\mathbb{T}}$ is the min-plus tropicalisation of Φ (minimum over its monomials);

(b) on positive rigid points, if Φ has nonnegative coefficients, then $\tau \circ \Phi^{\mathrm{an}} = \Phi_{\mathbb{T}} \circ \tau$;

(c) on arbitrary rigid points, $\tau \circ \Phi^{\mathrm{an}} \geq \Phi_{\mathbb{T}} \circ \tau$ coordinatewise, with strict inequality exactly when leading terms cancel.

In the diagram below, the outer square commutes on positive points by (b), and the path $\tau \circ \Phi^{\mathrm{an}} \circ s$ out of the skeleton equals $\Phi_{\mathbb{T}}$ everywhere by (a):

$$\begin{array}{ccc} F_\rho^{(k)} \subseteq (\mathit{params})_K^{>0} & \xrightarrow{\ \Phi\ } & \mathrm{Mix}_k(V) \subseteq \mathsf{D}_K X \\ \Big\downarrow{\scriptstyle \mathit{rigid}} & & \Big\downarrow{\scriptstyle \mathit{rigid}} \\ (\mathbb{A}^m)^{\mathrm{an}} & \xrightarrow{\ \Phi^{\mathrm{an}}\ } & (\mathbb{A}^n)^{\mathrm{an}} \\ {\scriptstyle s}\big(\big\uparrow{\scriptstyle \tau}\big\downarrow & & \Big\downarrow{\scriptstyle \tau} \\ F_{\mathbb{T},v}^{(k)} \subseteq \mathbb{R}^m & \xrightarrow[\ \Phi_{\mathbb{T}}\ ]{} & \mathrm{Mix}_k^{\mathbb{T}} \subseteq \mathsf{D}_{\mathbb{T}} X \end{array}$$

Proof. (a) $\tau_j(\Phi^{\mathrm{an}}(s(v))) = -\log|z_j|_{\Phi^{\mathrm{an}} s(v)} = -\log|\Phi_j|_{s(v)} = \min_m \langle m, v\rangle$ over the monomials of Φ_j, because nonzero integers have absolute value 1 when the residue characteristic is 0. (b) This is (1). (c) This is the ultrametric inequality $|f + g| \leq \max(|f|, |g|)$. □

Check T4 computes $-\log$ of the Gauss norms of the mixture coordinates symbolically at random skeleton points and recovers $\mu_{\mathbb{T}}$.

What this replaces. The withdrawn [1, §8.2] construction tensored $C(\Delta_X)$ with k. Here the algebra is $K[z]$, the polynomial functions in probability coordinates, whose analytification is standard. The simplex enters through positivity and the normalisation min = 0, not through the algebra. The Gel'fand topology of the old proposal becomes the Berkovich topology. The old proposal's "W_1 compatibility" has no counterpart here: τ is continuous by construction, and no metric is needed.

6 Shadows at zero temperature, and walls

Proposition 6.1 (The posterior tropicalises to the hard assignment)**.** *For $\rho = \sum_i w_i r_i$ positive, the Bayes posterior of the component given x, $P(i \mid x) = w_i r_i(x)/\rho(x)$, has*

$$\mathrm{val}\, P(i \mid x) = w_i^{\mathbb{T}} + u_i(x) - v(x) \geq 0,$$

with equality iff i attains the minimum. For convergent series its $t \to 0$ limit is $\mathrm{lc}(w_i r_i(x))/\mathrm{lc}(\rho(x))$ on the attainers and 0 elsewhere, where lc *denotes the leading coefficient. So the conditional expectation onto the observation, $\mathbb{E}[\,\cdot \mid x]$, tropicalises to the argmin assignment, with ties split by leading coefficients.*

Proof. (1) and the definition of the attainers (check T5). □

Remark 6.2 (What the earlier scripts were computing)**.** The scripts audited in [1, §9.2] assign each point to its nearest atom and average within groups. By proposition 6.1, the canonical version of that operation, conditioning on the observation in the star space [1, Def. 4.6], becomes exactly such a hard assignment at zero temperature, and their Voronoi walls are tie loci. The same mechanism underlies the small-variance limit of Gaussian mixtures, where EM becomes k-means and the walls become Voronoi or power diagrams [8]. The scripts therefore computed a tropical shadow while comparing it with classical theorems. This is a second reason, beyond those listed in [1], why their interchange counts carried no information about [1, Thm 3.3].

Proposition 6.3 (Walls)**.** *(a) On each assignment cell, $\mu_{\mathbb{T}}$ is affine. Crossing a tie hyperplane $w_i + u_i(x) = w_j + u_j(x)$ changes the assignment type and the affine piece (check T8). These hyperplanes are the tropical walls of the fibre and of the secant.*

(b) Component collision: if $r_i = r_j$, the classical fibre contains the weight-split segment $\{(w_i, w_j) : w_i + w_j = c\}$. Its valuation image is the pair of rays $\{\min(w_i^{\mathbb{T}}, w_j^{\mathbb{T}}) = \operatorname{val} c\}$, a one-dimensional tropical fibre. The collision locus Sec_{k-1} inside Sec_k (the correct "wall" of [1, §8.3]) tropicalises to the locus $u_i = u_j$, where the tropical fibre gains this bent line.

Proof. (a) A minimum of affine functions is affine wherever the set of attainers is constant. (b) By (1), $\operatorname{val}(w_i + w_j) = \min(\operatorname{val} w_i, \operatorname{val} w_j)$, and every pair with the stated minimum lifts, by the argument of theorem 3.2. □

7 What is established and what is open

Statement	Status
val is a monad morphism $\mathsf{D}_K \Rightarrow \mathsf{D}_{\mathbb{T}}$; the tower tropicalises	proved (theorem 2.2 and corollary 2.3)
classical fibres map onto tropical fibres	proved (theorem 3.2)
positive mixtures tropicalise to tropical secants, exactly	proved (theorem 4.2)
Berkovich square on the skeleton and on positive points	proved (theorem 5.1)
posterior → argmin; nearest-atom pooling is a tropical shadow	proved (proposition 6.1)
walls = tie hyperplanes; collision → bent line	proved (proposition 6.3)
lifting of tropical fibres *inside a model* V (tropical vs classical identifiability)	open (open problem 7.1)
tropical convex order on $F_{\mathbb{T},v}$ and the image of $\leq_{\mathrm{cx}}$	open (open problem 7.2)
Gaussian (non-finite) mixtures beyond the small-variance limit	open
Beck defects as Čech classes on the skeleton	not addressed; no construction is proposed

Open problem 7.1 (Tropical identifiability)**.** For $\rho \in \mathrm{Mix}_k(V)$, is $\operatorname{val}\left(F_\rho^{(k)} \cap D_k(V)\right) = F^{(k)}_{\mathbb{T},\operatorname{val}\rho} \cap D_k^{\mathbb{T}}(\mathrm{trop}^+ V)$? For $V = \mathsf{D}_K X$ this is theorem 3.2. In general, a tropical decomposition into model rates may fail to lift into the model. The failure locus would be a tropical certificate of non-identifiability. Whether it detects finite versus positive-dimensional classical fibres is the identifiability question, and example 4.4 gives the first data.

Open problem 7.2 (Order on tropical fibres)**.** Corollary 2.3 sends classical poolings to tropical poolings, so the image of the convex order is contained in the transitive closure of tropical pooling. Is that closure a partial order on $F_{\mathbb{T},v}$ with bottom and top the two points of proposition 3.3, and does it orient the tropical Beck defects of the shift law as [1, Prop. 5.4] does classically?

Code. `check_tropical.py`: 8 checks. T1–T5 and T8 are exact rational arithmetic on real Puiseux series; T6–T7 are floating-point Jacobian ranks. All pass.

References

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